Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The point from where a ball is projected is taken as the origin of the coordinate axes. The x and y components of its displacement are given by $x = 6t$ and $y = 8t - 5t^2$, What is the velocity of projection?

(A) $6\ \mathrm{m/s}$

(B) $8\ \mathrm{m/s}$

(C) $10\ \mathrm{m/s}$

(D) $14\ \mathrm{m/s}$

Q.2

Three particles A, B and C are situated at the vertices of an equilateral triangle of side l at $t = 0$. The particle A heads towards B, B towards C, C towards A with constant speeds v. Find the time of their meeting.

(A) $\frac{2l}{3v}$

(B) $\frac{3l}{2v}$

(C) 2l

(D) $\frac{3l}{4v}$

Q.3

A particle is projected with a velocity v so that its range on a horizontal plane is twice the greatest height attained If g is acceleration due to gravity, then its range is

(A) $\frac{4v^2}{5g}$

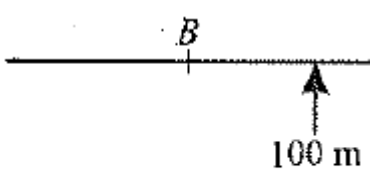
(B) $\frac{4g}{5v^2}$

(C) $\frac{4v^3}{5g^2}$

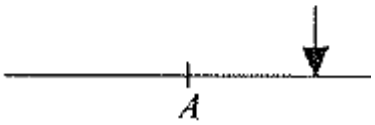
(D) $\frac{4v}{5g^2}$

Q.4

A man swims from a point A on one bank of a river of width $100\ \mathrm{m}$. When he swims perpendicular to the water current, he reaches the other bank $50\ \mathrm{m}$ downstream. The angle to the bank at which he should swim, to reach the directly opposite point B on the other bank is



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(A) 10° upstream

(B) 20° upstream

(C) 30° upstream

(D) 60° upstream

Q.5

A body is moving in a circular path with a constant speed It has

(A) a constant velocity.

(B) a constant acceleration.

(C) an acceleration of constant magnitude.

(D) an acceleration which varies with time in magnitude.

Q.6

A ball thrown by one player reaches the other in 2 s. The maximum height attained by the ball above the point of projection will be about

(A) 2.5 m

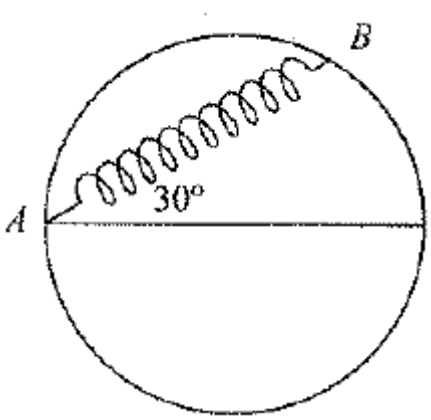
(B) 5 m

(C) 7.5 m

(D) 10 m

Q.7

A bead of mass m is attached to one end of a spring of natural length R and spring constant $K = \frac{(\sqrt{3}+1)mg}{R}$ The other end of the spring is fixed at a point A on a smooth vertical ring of radius R as shown in the Fig. 7.324. The normal reaction at B just after it is released to move is



(A) $mg/2$

(B) $\sqrt{3}mg$

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(C) $3\sqrt{3}mg$

(D) $\frac{3\sqrt{3}mg}{2}$

Q.8

A particle is projected with a certain velocity at an angle α above the horizontal from the foot of an inclined plane of inclination 30°. If the particle strikes the plane normally then α is equal to

(A) $30^\circ + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(B) 45°

(C) 60°

(D) $30^\circ + \tan^{-1}(2\sqrt{3})$

Q.9

A projectile is thrown in the upward direction making an angle of 60° with the horizontal direction with a velocity of $150\ \text{ms}^{-1}$. Then the time after which its inclination with the horizontal is 45°, is

(A) $15(\sqrt{3}-1)\text{s}$

(B) $15(\sqrt{3}+1)s$

(C) $7.5(\sqrt{3}-1)\text{s}$

(D) $7.5(\sqrt{3}+1)\text{s}$

Q.10

Find the time period of meeting of minute hand and second hand of a clock.



(A) 58/60

(B) 59/60

(C) 60/59

(D) 58/9

Q.11

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The car is moving towards cast with a speed of $25\ \mathrm{km/h}$. To the driver of the car, a bus appears to move towards north with a speed of $25\sqrt{3}\ \mathrm{km/h}$. What is the actual velocity of the bus?

(A) $50\ \mathrm{km/h}, 30^\circ$ east of north

(B) $50\ \mathrm{km/h}, 30^\circ$ north of east

(C) $25\ \mathrm{km/h}, 30^\circ$ east of north

(D) $25\ \mathrm{km/h}, 30^\circ$ north of east

Q.12

The angular velocity of a particle moving in a circlc of radius $50\ \mathrm{cm}$ is increased in 5 min from 100 revolutions per minute to 400 revolutions per minute. Find tangential acceleration of the particle.

(A) $60\ \mathrm{m/s^2}$

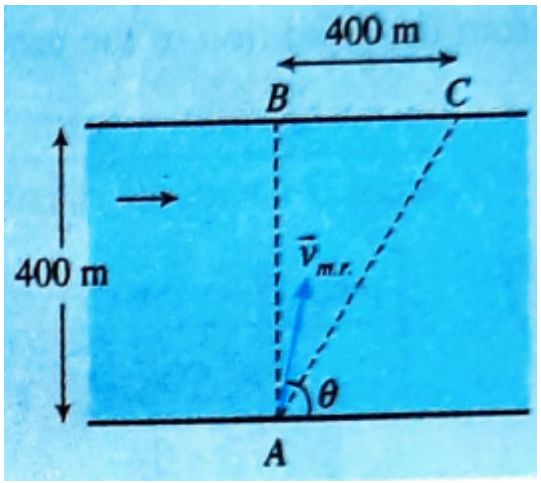
(B) $\pi/30\ \mathrm{m/s^2}$

(C) $\pi/15\ \mathrm{m/s^2}$

(D) $\pi/60\ \mathrm{m/s^2}$

Q.13

A river is flowing with a speed of $1\mathrm{kmh}^{-1}$. A swimmer wants to go to point C starting from A. He swims with a speed of 5 kmh^{-1} at an angle θ w.r.t. the river flow. If $AB = BC = 400\ \mathrm{m}$, at what angle with the river bank should the swimmer swim?



(A) 37

(B) 53

(C) 47

(D) 51

Q.14

A plank is moving on a ground with a velocity v and a block is moving on the plank with a velocity u as shown in figure. What is the velocity of block with respect to ground?

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(A) $v - u$ towards right

(B) $v - u$ towards left

(C) u towards right

(D) None of these

Q.15

A particle is moving in a circle of radius r centred at O with a constant speed v. What is the change in the velocity in moving from A to B $(\angle AOB = 40^\circ)$?

(A) $2v \sin 20^\circ$

(B) $4v \sin 40^\circ$

(C) $2v \sin 40^\circ$

(D) $v \sin 20^\circ$

Q.16

A body of mass m is projected horizontally with a velocity. v from the top of a tower of height h and it reaches the ground at a distance x from the foot of the tower. If a second body of mass $2\ \mathrm{m}$ is projected horizontally from the top of a tower of height $2h$, it reaches the ground at a distance $2x$ from the foot of the tower. The horizontal velocity of the second body is

(A) v

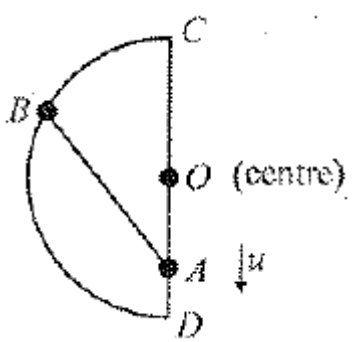
(B) $2v$

(C) $\sqrt{2}v$

(D) $v/2$

Q.17

Two beads A and B move along a semicircular wire frame as shown in Fig. The beads are connected by an inelastic string with always remains tight. At as instant the speed of A is u, $\angle BAC = 45^\circ$ and $\angle BOC = 75^\circ$, where O is the centre of the semicircular ar(c) The speed of bead B at that instant is



(A) $\sqrt{2}u$

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(B) u

(C) $\frac{u}{2\sqrt{2}}$

(D) $\sqrt{\frac{2}{3}}u$

Q.18

An air craft executes a horizontal loop of radius 1kn with a steady speed of 900kmh^{-1}. Find its centripetal acceleration.

(A) $62.5\ \text{m/s}^2$

(B) $30\ \text{m/s}^2$

(C) $32.5\ \text{m/s}^2$

(D) $72.5\ \text{m/s}^2$

Q.19

A policeman moving on a highway with a speed of 30 kmh^{-1} fires a bullet at a thief's car speeding away in the same direction with a speed of 192kmh^{-1}. If the muzzle speed of the bullet is $150\ \text{ms}^{-1}$ with what speed does the bulet hit the thief's car?

(A) $120\ \text{m/s}$

(B) $90\ \text{m/s}$

(C) $125\ \text{m/s}$

(D) $105\ \text{m/s}$

Q.20

Shots are fired simultaneously from the top and bottom of a vertical cliff with the elevation $\alpha = 30°, \beta = 60°$, respectively (Fig. A.15). The shots strike an object simultaneously at the same point. If $\text{a} = 30\sqrt{3}\ \text{m}$ is the horizontal distance of the object from the cliff, then the height h of the cliff is

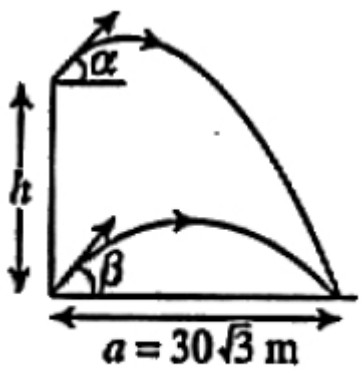


Fig. A.15

(A) $30\ \text{m}$

(B) $45\ \text{m}$

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(C) $60\ \mathrm{m}$

(D) $90\ \mathrm{m}$

Q.21

A boat is sent across (perpendicular) a river with a velocity of $8\mathrm{kmh}^{-1}$. If the resultant velocity of the boat is $10\mathrm{kmh}^{-1}$, the river is flowing with a velocity

Q.22

A man is sitting in an open car which is travelling along a road at a speed of $30\ \mathrm{m\ s}^{-1}$. The man stands up and throws a ball at a speed of $15\ \mathrm{m\ s}^{-1}$ relative to himself at an angle θ to the horizontal. The moment the man throws the ball, the driver of the car begins to accelerate at a constant rate of $10\sqrt{3}\ \mathrm{m\ s}^{2}$, in the direction in which it was initially travelling. Find the time at which the man catches the ball if the instant at which the ball was thrown is $t = 0$.
Assume that the time at which the ball travels contains the line of the motion of the car, and that $g = 10\ \mathrm{m\ s}^{-2}$.
(mark the nearest integer of time at which the man catches the ball if the instant at which the ball was thrown is t=0)
(take angle θ as 30°)

Q.23

A particle is moving along a circular path with uniform speed. Through what angle does its angular velocity change when it completes half of the circular path?

Q.24

The front windscreen of a car is inclined at an angle 60° with the vertical. Halstones fall vertically downwards with a speed of $5\sqrt{3}\ \mathrm{m/s}$. Find the speed of the car so that hailstones are bounced back by the sereen in vertically upward direction with respect to car Assume elastic collision of haistones with wind screen. Find the speed of car divided by 5.

Q.25

A golfer standing on level ground hits a ball with a velocity of $u = 52\ \mathrm{m/s}$ at an angle a above the horizontal. If $\tan a = 5/12$, then the time for which the ball is at least 15 m above the ground will be (take $g = 10\ \mathrm{m/s^2}$)

Q.26

A ball thrown by one player reaches the other in $2\ \mathrm{s}$. The maximum height attained by the ball above the point of projection will be about

Q.27

Rain is falling vertically downwards with a speed of $4\mathrm{kmh}^{-1}$. A girl moves on a straight road with a velocity of $3\mathrm{kmh}^{-1}$. The apparent velocity of rain with respect to the girl is

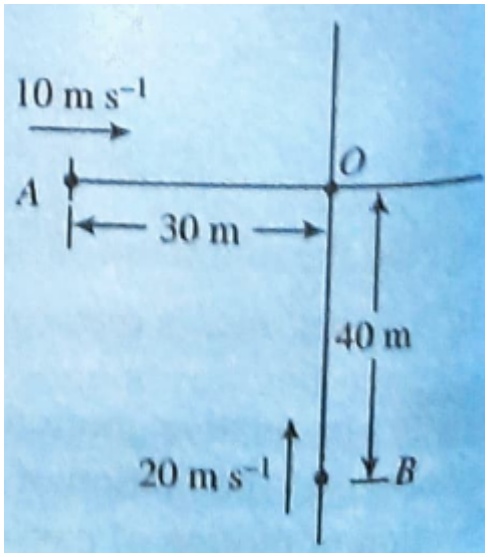
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Q.28

Two cars A and B are moving west to east and south to north, respectively, along crossroads. A moves with a speed of 20 ms^{-1} and is $500\ \mathrm{m}$ away from the point of intersection of cross roads and B moves with a speed of $15\ \mathrm{m\ s}^{-1}$ and is $400\ \mathrm{m}$ away from the point of intersection of cross roads. Find the shortest distance between them. Find shortest distance/5.

Q.29

Two particles A and B are moving with uniform velocity as shown in figure given below at $t = 0$. Find out shortest distance is $x\sqrt{5}$ than x will?



Q.30

A ball is thrown with a velocity whose horizontal component is $12\ \mathrm{ms}^{-1}$ from a point $15\ \mathrm{m}$ above the ground and $6\ \mathrm{m}$ away from a vertical wall $18.75\ \mathrm{m}$ high in such a way to just clear the wall. At what time will the ball reach the ground?(take $g = 10\ \mathrm{ms}^{-2}$)

Answers & Solutions

Q.1 Answer:

$10\ \mathrm{m/s}$

Solution:

$v_x = \frac{dx}{dt} = 6$ and $v_y = \frac{dy}{dt} = 8 - 10t$

put $t = 0$($\because$ we have to find initial velocity)

$v_y = 8 - 10 \times 0 = 8$

$v = \sqrt{v_x^2 + v_y^2} = \sqrt{6^2 + 8^2} = 10\ \mathrm{m/s}$

Q.2 Answer:

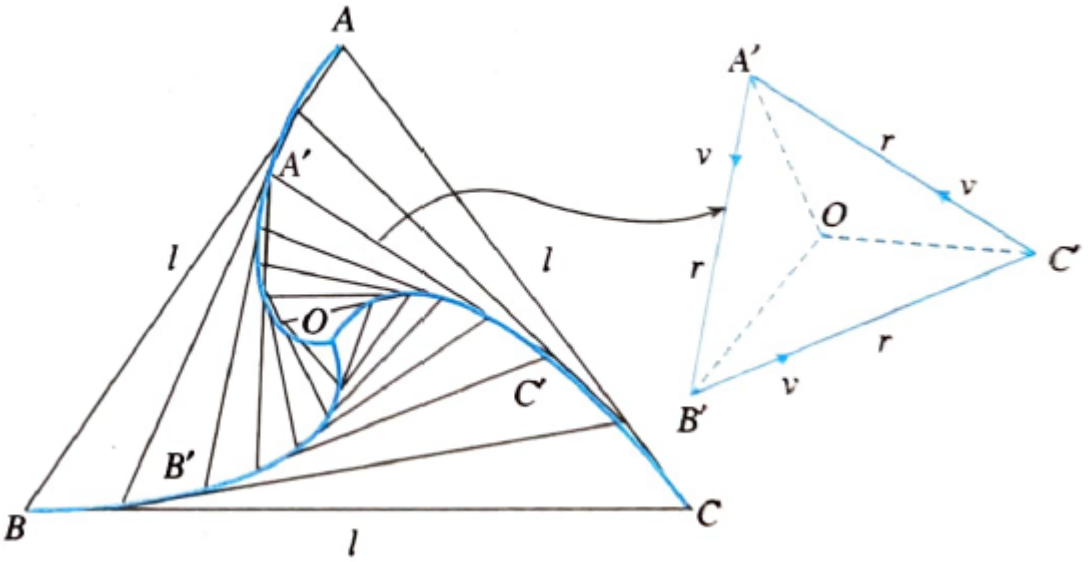
$\frac{2l}{3v}$

Solution:

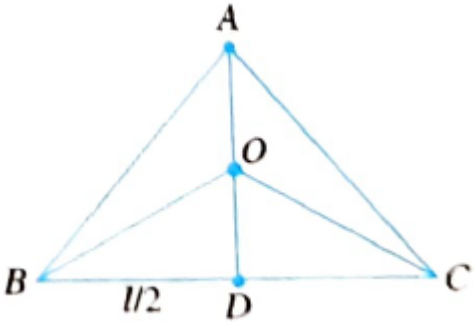
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The motion of the particles is roughly sketched in figure. By symmetry they will meet at the centroid O of the triangle. As the speed of all the particles is equal they will cover equal distance in any given interval of time.

If we join the instantaneous position of the particle at any time, the particles will form an equilateral triangle of same centroid as initial triangle.



Let us consider the motion of any one particle say A. At any instant, its velocity makes an angle 30° with line AO.



The component of the velocity along AO is $v\cos 30^\circ$. This component will be equal to the rate of change of distance between A and O.

At $t = 0$, distance between A and O

$$AO = \frac{2}{3}AD = \frac{2}{3}\sqrt{l^2 - \left(\frac{l}{2}\right)^2} = \frac{l}{\sqrt{3}}$$

At time $t = T$, the separation between A and O is zero. Hence, time taken for AO to become zero,

$$T = \frac{(l/\sqrt{3})}{v\cos 30^\circ} = \frac{(l/\sqrt{3})}{v(\sqrt{3}/2)} = \frac{2l}{3v}$$

Q.3 Answer:

$\frac{4v^2}{5g}$

Solution:

Since $R = 2H$ or $\frac{v^2\sin 2\theta}{g} = 2 \times \frac{v^2\sin^2\theta}{2g}$ or $2\sin\theta\cos\theta = \sin^2\theta$ or $\tan\theta = 2$

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$$R = \frac{v^2 \sin 2\theta}{g} = \frac{v^2 2 \sin\theta \cos\theta}{g} = \frac{2v^2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4v^2}{5g}$$

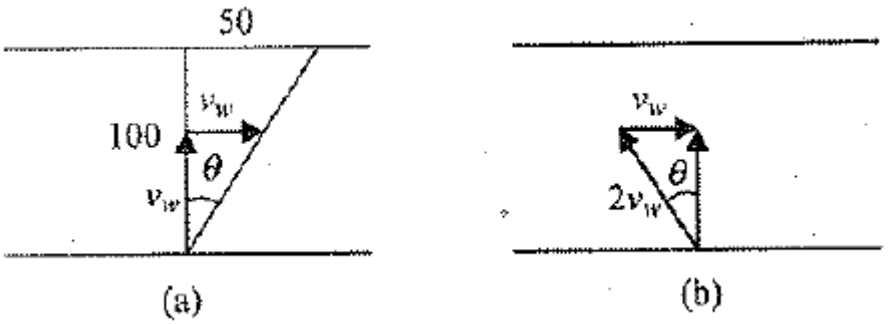
Q.4 Answer:

60° upstream

Solution:

Refer to Fig.

$\tan\theta = \frac{v_w}{v_m} = \frac{50}{100} = \frac{1}{2}$ or $v_m = 2v_w$



Refer to Fig.

$\sin\theta = \frac{v_w}{v_m} = \frac{v_w}{2v_{wy}} = \frac{1}{2}$ **or** $\theta = 30^\circ$

So, it is 60° upstream.

Q.5 Answer:

an acceleration of constant magnitude.

Solution:

$a_c = \frac{v^2}{r} \rightarrow$ **constant in magnitude if v is constant.**

Q.6 Answer:

5 m

Solution:

$t = \frac{2u\sin\theta}{g}$ **or** $2 = \frac{2u\sin\theta}{g} \Rightarrow u\sin\theta = g$

$h_m = \frac{u^2\sin^2\theta}{2g} = \frac{g^2}{2g} = \frac{g}{2} = 5$ m

Q.7 Answer:

$\frac{3\sqrt{3}mg}{2}$

Solution:

Extension in the spring is

$$x = AB - R$$

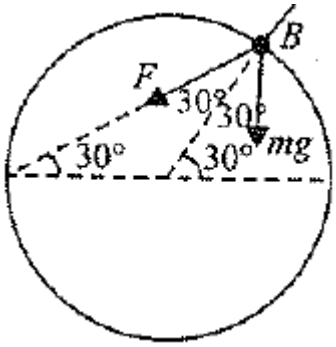
$$= 2R\cos 30^\circ - R = (\sqrt{3} - 1)R$$

Spring force

$F = kx = \frac{(\sqrt{3}+1)mg}{R} \times (\sqrt{3} - 1)R = 2mg$

N

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From Fig.

$N = (F + mg)\cos 30^\circ = \frac{3\sqrt{3}mg}{2}$

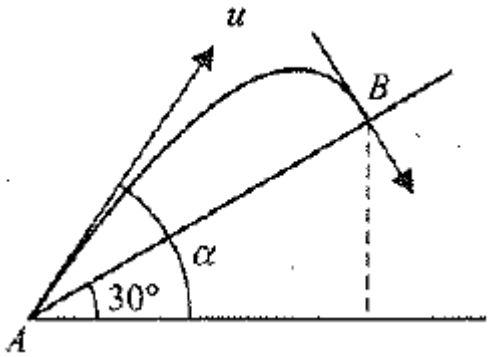
Q.8 Answer:

$30^\circ + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Solution:

t_{AB} **= time of flight of projectile** $= \frac{2u\sin(\alpha - 30^\circ)}{g\cos 30^\circ}$

Now component of velocity along the plane becomes zero at point B



$0 = u\cos(\alpha - 30^\circ) - g\sin 30^\circ \times T$

or $u\cos(\alpha - 30^\circ) = g\sin 30^\circ \times \frac{2u\sin(\alpha - 30^\circ)}{g\cos 30^\circ}$ **or** $\tan(\alpha - 30^\circ) = \frac{\cot 30^\circ}{2} = \frac{\sqrt{3}}{2}$

or $\alpha = 30^\circ + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Q.9 Answer:

$7.5(\sqrt{3} - 1)\text{s}$

Solution:

At the two points of the trajectory during projectile motion, the horizontal component of the velocity is same. Then

$u\cos 60^\circ = v\cos 45^\circ$

$\Rightarrow \quad 150 \times \frac{1}{2} = v \times \frac{1}{\sqrt{2}}$ or $v = \frac{150}{\sqrt{2}}$ m/s

Initially : $u_y = u\sin 60^\circ = \frac{150\sqrt{3}}{2}$ m/s

Finally : $v_y = \text{vsin}45^\circ = \frac{150}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{150}{2}$ m/s

But $v_y = u_y + a_y t$ **or** $\frac{150}{2} = \frac{150\sqrt{3}}{2} - 10t$

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$10t = \frac{150}{2}(\sqrt{3}-1)$ **or** $t = 7.5(\sqrt{3}-1)\text{s}$

Q.10 Answer:

60/59

Solution:

$\omega_{\text{min}} = \frac{2\pi}{60}\text{rad/ min}$ **and** $\omega_{\text{sec}} = \frac{2\pi}{1}\text{rad/min}$

The rate of change of angle between minute hand and second hand is $\frac{d\theta}{dt} = \omega_{\text{sec}} - \omega_{\text{min}}$

$(\omega_{\text{se}} - \omega_{\text{min}}) \cdot T = 2\pi$

$\left(\frac{2\pi}{1} - \frac{2\pi}{60}\right) T = 2\pi \Rightarrow T = \frac{1}{\left(1-\frac{1}{60}\right)} = \frac{60}{59}\,\text{min}$

Q.11 Answer:

$50\ \text{km/h}, 30^\circ$ east of north

Solution:

V_b = **velocity of bus**

V_c = **velocity of car**

$\text{V}_{\text{bc}} = 25\sqrt{3}\ \text{km/hr}$

now from relative velocity

$\left|\overrightarrow{\text{V}}_\text{b}\right| = \sqrt{(25)^2 + (25\sqrt{3})^2}$

$= 50\ \text{km/hr}$

for direction

$\tan\theta = \frac{25\sqrt{3}}{25} = \sqrt{3}$

so ' θ **' is from east direction**

Therefore it will be at $30'$ **east of north**

Hence (A) is correct

Q.12 Answer:

$\pi/60\ \text{m/s}^2$

Solution:

$\alpha = (\omega_2 - \omega_1)/t = (400-100)/5 = 60\text{rev/min}^2$

$= \frac{60\times 2\pi}{(60)^2} = \frac{2\pi}{60}\text{rad/s}^2$

$a_f = \alpha r = \frac{2\pi}{60} \times \frac{50}{100} = \frac{\pi}{60}\ \text{m/s}^2$

Q.13 Answer:

53

Solution:

Resultant path of the swimmer is at 45° **with bank, therefore,** x **- and** y**-components of swimmer's resultant velocity must be equal.**

Let velocity of swimmer $|\vec{v}_m| = v$

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$\vec{v}_m = \vec{v}_{m,r} + \vec{v}_r$

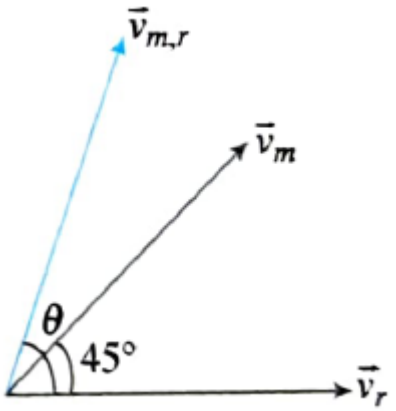
$v_x = v_r + v_{m,r}\cos\theta$

$v_y = v_{m,r}\sin\theta$

On squaring, $1 + 25\cos^2\theta + 10\cos\theta = 25 - 25\cos^2\theta$

$50\cos^2\theta + 10\cos\theta - 24 = 0$

On solving, we get $\theta = 53^\circ$.



Q.14 Answer:

$v - u$ towards right

Solution:

$\vec{v}_p = v\hat{i}, \vec{v}_{b/p} = -u\hat{i}$

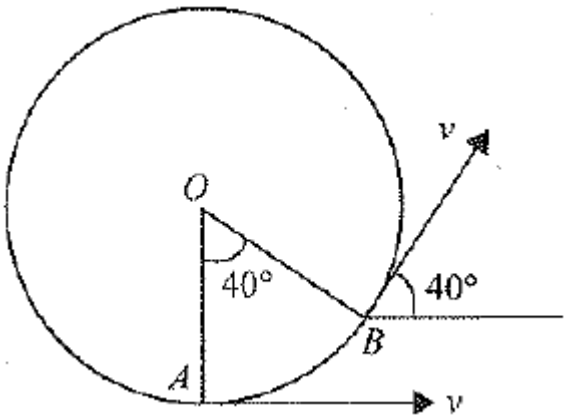
$\vec{v}_b = \vec{v}_p + \vec{v}_{b/p} = (v-u)\hat{i} = v - u$ **towards right.**

Q.15 Answer:

$2v\sin 20^\circ$

Solution:

Change in velocity $= \vec{v}_f - \vec{v}_i$



Its magnitude is $\sqrt{v^2 + v^2 - 2vv\cos 40^\circ} = 2v\sin 20^\circ$

Q.16 Answer:

$\sqrt{2}v$

Solution:

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$x = v\sqrt{\frac{2h}{g}}, 2x = v'\sqrt{\frac{2(2h)}{g}}$

Solve to get: $v' = \sqrt{2}v$

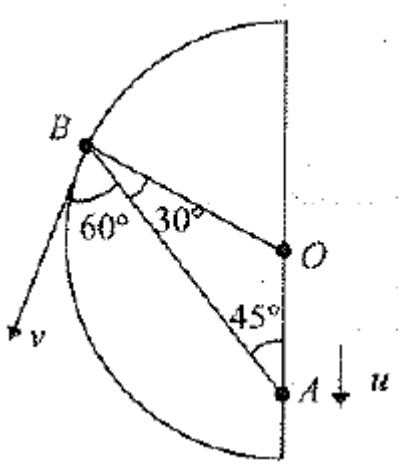
Q.17 Answer:

$\sqrt{2}u$

Solution:

As in Fig.

$u\cos 45^\circ = v\cos 60^\circ$



or $v = \sqrt{2}u$

Q.18 Answer:

$62.5\ \text{m/s}^2$

Solution:

$r = 1\ \text{km} = 1000\ \text{m}, v = 900\ \text{km/h} = 900 \times 5/18 = 250\ \text{m/s}$

$$a_c = \frac{v^2}{r} = \frac{(250)^2}{1000} = 62.5\ \text{m/s}^2$$

Q.19 Answer:

$105\ \text{m/s}$

Solution:

Velocity of police van $= 30 \times \frac{5}{18} = \frac{25}{3}\ \text{m/s}$

Muzzle speed of the bullet $= 1.50\ \text{ms}^{-1}$

Speed of the bullet w.r.t ground $= (150 + 25/3)\text{ms}^{-1}$

Velocity of thief's car

$= 192 \times \frac{5}{18} = \frac{32\times 5}{3} = \frac{160}{3}\ \text{m/s}$

∴ relative velocity of the bullet w.r.t. the thief's car

$= 150 + \frac{25}{3} - \frac{160}{3} = 150 - \frac{135}{3} = 105\ \text{m/s}$

Q.20 Answer:

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60 m

Solution:

Given,

$\alpha = 30^\circ, \quad \beta = 60^\circ$

$a = 30\sqrt{3}\ \text{m}$

Let the velucity be V and u.

$\text{Range } = \frac{u^2 \sin 2\theta}{g}$

So, for first shot

$R = 30\sqrt{3}\ \text{m} = \frac{\text{v}^2 \sin 2\times 60^\circ}{10}$

$v = 10\sqrt{6}\ \text{m/sec}$

Also,

$s = vt \quad (\text{ in horizontal aistance })$

$$30\sqrt{3} = u\cos\beta t$$

$$t = \frac{30\sqrt{3}}{1016 \times \cos 60^\circ}$$

$$t = 3\sqrt{2}\text{sec}$$

second shot also moves same distunce in time t. so,

$v\cos\alpha \times t = 30\sqrt{3}$

$v\cos 30^\circ \times 3\sqrt{2} = 30\sqrt{3}$

$v = \frac{30\sqrt{3}\times 2}{\sqrt{3}\times 3\sqrt{2}}$

$v = 10\sqrt{2}\ \text{m/sec}$

vertically H distence is covered in timet.

$$S_y = u_y t + \frac{1}{2}at^2$$

$$-H = v\sin 30^\circ \times 3\sqrt{2} - \frac{1}{2} \times 9 \times (3\sqrt{2})^2$$

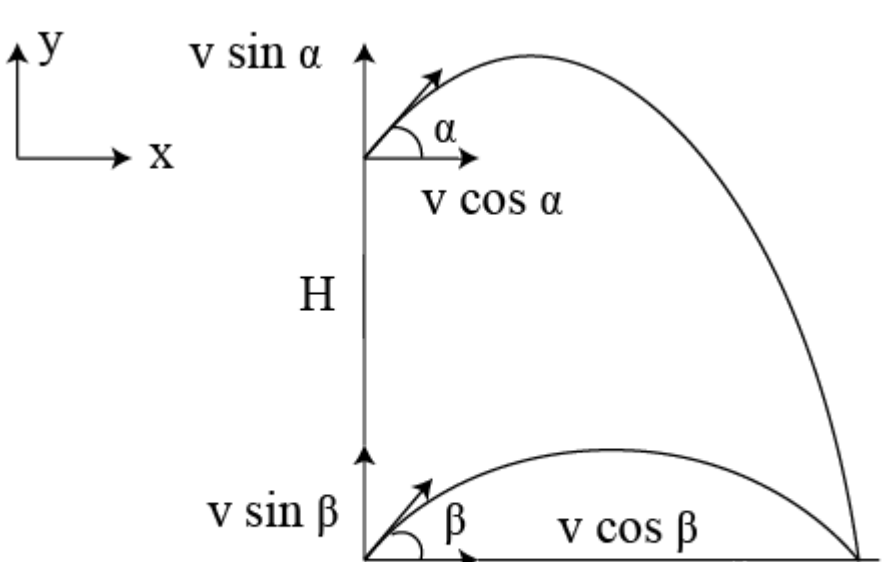
$$-H = 5\sqrt{2} \times 312 - 90$$

$$-H = 30 - 90$$

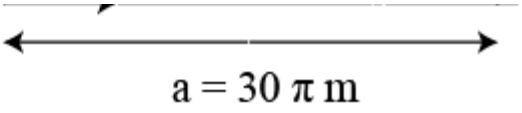
$$-H = -60$$

$$H = 60\ \text{m}$$

So, the heightis 60 m



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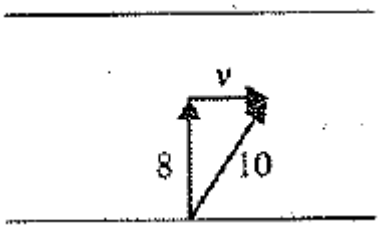


Q.21 Answer:

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Solution:

$10^2 = v^2 + 8^2$ **(Fig.)**



or $v^2 = 100 - 64 = 36$ **or** $v = 6\text{kmh}^{-1}$

Q.22 Answer:

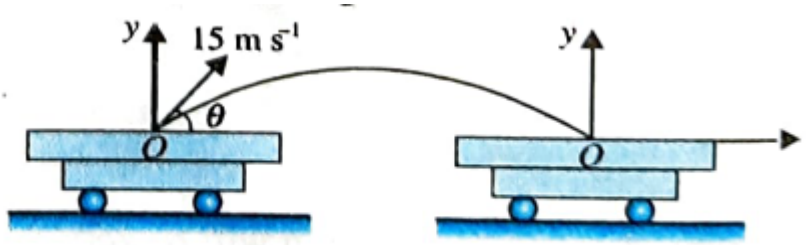
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Solution:

You can attempt the problem in a reference frame fixed to the car or in the reference frame fixed to the ground. Fundamentally, these two reference frames are equivalent in all respects, but in the reference frame fixed to the car, the calculations are slightly easier.

Analysis of the motion of the ball in the reference frame fixed to the car: Let x**-axis of the coordinate system be along the motion of the car, and** y**-axis along the vertical direction. Acceleration of the ball in the reference frame,**

$\vec{a}_{\text{BR}}$ **= Acceleration of the ball w.r.t. ground** $-$ **Acceleration of the reference frame w.r.t. ground**



In time interval $t = 0$ to $t = t$, $x = 0$, $y = 0$

$$\vec{a}_{BR} = (-g\hat{j}) - (a\hat{i}) = -a\hat{i} - g\hat{j}$$

$$\Rightarrow \quad (a_{BR})_x = -a, (a_{BR})_y = -g$$

Let u **be the velocity of the ball at** $t = 0$**. Then** $u_x = u\cos\theta, u_y = u\sin\theta$

Using the equation, $s = ut + \frac{1}{2}at^2$**, in both** x **- and** y**-directions,**

$0 = (u\cos\theta)t + \frac{1}{2}(-a)t^2$ **...(i)**

$\text{At} t = t, x = 0, y = 0$

and $0 = (u\sin\theta)t + \frac{1}{2}(-g)t^2$**...(ii)**

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The man catches the ball.

From these equations, either $t = 0$, which refers to the initial condition, or $t = \frac{2u\sin\theta}{g}$ and $t = \frac{2u\cos\theta}{a}$.

$$\frac{2u\sin\theta}{g} = \frac{2u\cos\theta}{a}$$

$$\tan\theta = \frac{g}{a} = \frac{10}{10\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \quad \theta = 30^\circ$$

$$t = \frac{2u\sin\theta}{g} = \frac{2 \times 15 \times \frac{1}{2}}{10} = 1.5 \text{ s}$$

Q.23 Answer:

—

Solution:

Angular velocity is always directed perpendicular to the plane of the circular path. Hence, required change in angle $= 0^\circ$.

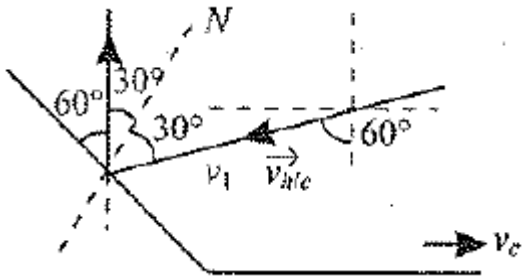
Q.24 Answer:

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Solution:

Let $\vec{v}_c = v_0\hat{i}, \vec{v}_{h/c} = -\frac{v_1}{2}\hat{j} - v_1\frac{\sqrt{3}}{2}\hat{i}$

$\vec{v}_h = -\frac{v_1}{2}\hat{j} + \left(v_0 - v_1\frac{\sqrt{3}}{2}\right)\hat{i}$



Now: $v_0 - v_1\frac{\sqrt{3}}{2} = 0 \Rightarrow v_1 = \frac{2v_0}{\sqrt{3}}$

also: $\left(\frac{v_1}{2}\right)^2 + \left(v_0 - v_1\frac{\sqrt{3}}{2}\right)^2 = (5\sqrt{3})^2 \Rightarrow v_1^2 = 75 \times 4 \Rightarrow \frac{4v_0^2}{3} = 75 \times 4 \Rightarrow \quad v_0 = 15 \text{ m/s}$

Q.25 Answer:

—

Solution:

Let at any time t, the ball is at height of 15 m.

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$S_y = u_y t + \frac{1}{2} a_y t^2$
$\Rightarrow 15 = u \sin\theta t - \frac{1}{2} g t^2$
$\Rightarrow 15 = 52 \times \frac{5}{13} t - \frac{1}{2} \times 10 t^2$
$\Rightarrow t^2 - 4t + 3 = 0 \Rightarrow (t-1)(t-3) = 0$
$\Rightarrow t = 1 \text{ s}, t = 3 \text{ s},$ required time: $3 - 1 = 2 \text{ s}$

Q.26 Answer:

—

Solution:

$t = \frac{2u \sin\theta}{g}$ **or** $2 = \frac{2u \sin\theta}{g}$ **or** $u \sin\theta = g$
$h_m = \frac{u^2 \sin^2\theta}{2g} = \frac{g^2}{2g} = \frac{g}{2} = 5 \text{ m}$

Q.27 Answer:

—

Solution:

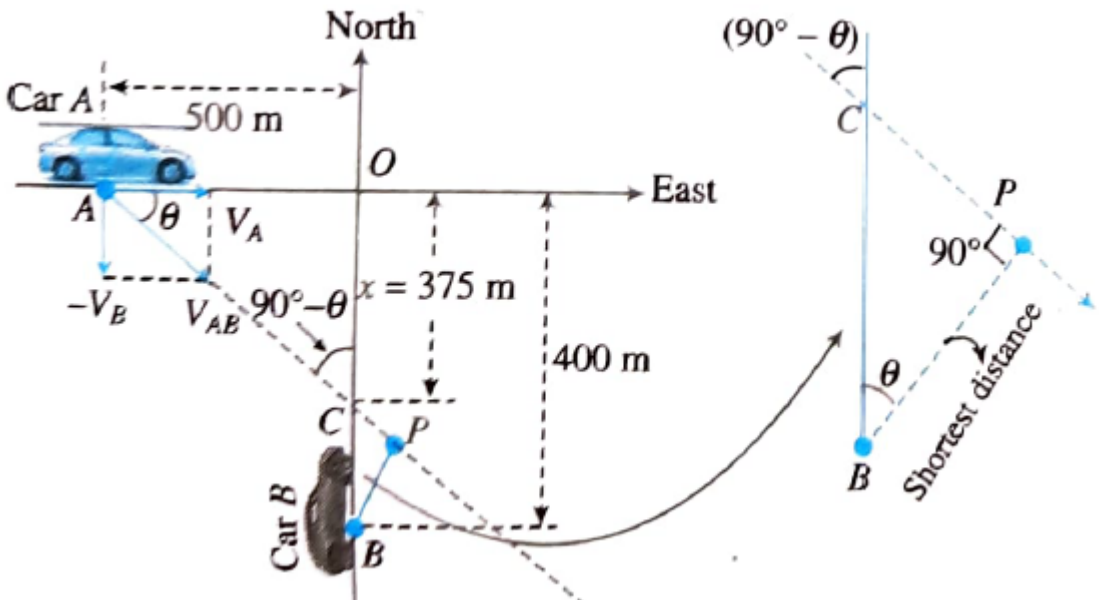
$\vec{v}_{r/g} = \vec{v}_r + (-\vec{v}_g)$
$v_{rg} = \sqrt{v_r^2 + v_g^2} = \sqrt{16+9} \text{ km h}^{-1} = 5 \text{ km h}^{-1}$

Q.28 Answer:

—

Solution:

(Using the concept of relative velocity) Let us assume the velocity of A w.r.t. B. To do this, we plot the resultant velocity, $\vec{V}_{AB}$.
$\bar{V}_{AB} = \bar{V}_A - \bar{V}_B = \bar{V}_A + (-\bar{V}_B)$
As the accelerations of both the cars is zero, so the relative acceleration between them is also zero. Hence, the relative velocity will remain constant. So the path of A with respect to B will be straight line and along the direction of relative velocity of A with respect to B. The shortest distance between A and B is a perpendicular from B on the line of motion of A with respect to B. From figure,
$\tan\theta = \frac{V_B}{V_A} = \frac{15}{20} = \frac{3}{4}$ **...(i)**



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This θ is the angle made by the resultant velocity vector $\left|\vec{V}_{AB}\right|$ with the x-axis. Let $OC = x$

Also we know that from figure, In $\triangle AOC$,

$\tan\theta = \frac{x}{500} = \frac{3}{4}$ **...(ii)**

From equation (i) and (ii), we get $x = 375$ m.

$\therefore \quad BC = OB - OC = 400 - 375 = 25$ m

But the shortest distance is BP

From diagram, it is clear that $BP = BC\cos\theta = 25 \times \frac{4}{5}$

$\therefore \quad BP = 20$ m

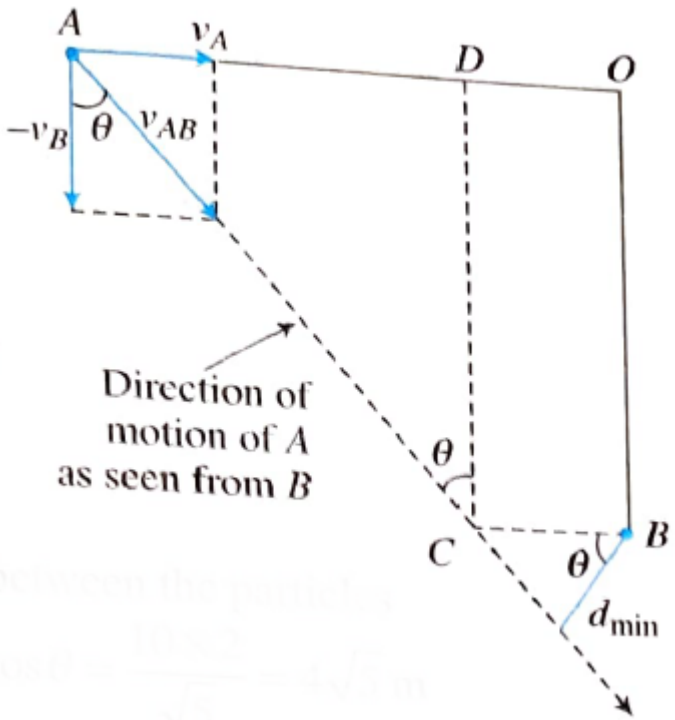
Q.29 Answer:

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Solution:

Observing A from the frame of B

Let us draw $\vec{v}_{AB}$,



$\tan\theta = \frac{10}{20} = \frac{1}{2}$

Again $\tan\theta = \frac{AD}{CD} = \frac{AD}{40} = \frac{1}{2}$

$\Rightarrow \quad AD = 20 \Rightarrow DO = 10$

$\Rightarrow \quad BC = 10$

Hence, the shortest distance between the particles

$d_{\text{short}} = BC\cos\theta = 10\cos\theta = \frac{10\times 2}{\sqrt{5}} = 4\sqrt{5}$ m

Since the closest distance is non-zero, therefore, the particles will not collide.

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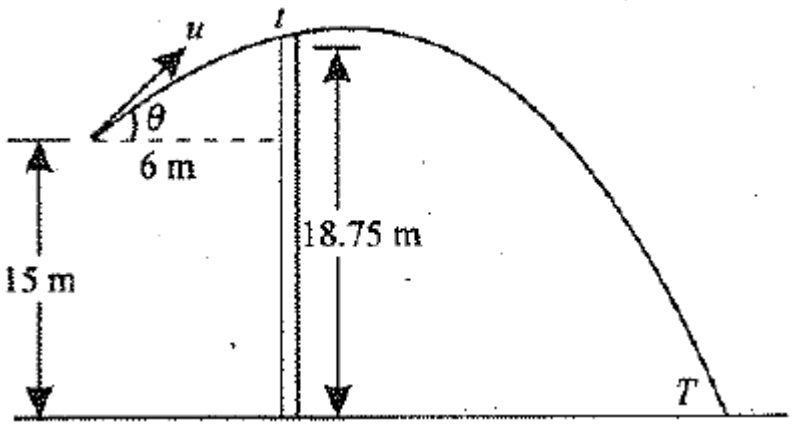
Q.30 Answer:

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Solution:

$u\cos\theta = 12, 6 = u\cos\theta t \Rightarrow t = 0.5$ s **(Fig.)**

$3.75 = u\sin\theta t - \frac{1}{2}gt^2$



$\Rightarrow \frac{15}{4} = u\sin\theta \times \frac{1}{2} - \frac{1}{2}10\left(\frac{1}{2}\right)^2$

$\Rightarrow u\sin\theta = 10$

$-15 = 10T - \frac{1}{2}10T^2 \Rightarrow T^2 - 2T - 3 = 0 \Rightarrow T = 3$ s

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